

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · MATHEMATICS

MATHEMATICS 4004/1

PAPER 1 November 2022

2 hours 30 minutes

PAPER 1 — SHORT-ANSWER — CALCULATORS MUST NOT BE USED

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2022 4004/1.

Additional materials: Electronic calculators must NOT be used. Geometrical instruments may be used.

INSTRUCTIONS: Answer all 30 questions. Show all working. The number of marks is given in brackets [].

1. Express 48 as a product of its prime factors. **[3]**

2. Expand and simplify $3(x + 1) - 3x$. **[2]**

3. A map of Bindura uses scale 2 : 800000. A road is 10 cm on the map. Find the real length in km. (1 : 100 000 means 1 cm $\leftrightarrow$ 1 km if 2:8 is first simplified — instead: scale 2 cm to 8 km.) **[3]**

4. Work out $\frac{3}{3} \div \frac{2}{3}$. **[3]**

5. In a class of Moleli, $n(A) = 12$ (play soccer), $n(A \cap B) = 3$ (play both). How many play soccer only? **[2]**

6. A school jersey costs \$250. The price rises by 25%. Find the new price. **[3]**

7. Write 90000 in standard form. **[2]**

8. Evaluate $-14 - (-12)$. **[2]**

- 9.** Simplify $3^3 \times 3^3$. Leave as a power of 3. **[3]**
- 10.** Find the HCF and LCM of 36 and 20. **[4]**
- 11.** Make r the subject of $C = 2\pi r$. **[4]**
- 12.** A sequence is 8, 12, 16, 20, ... Find an expression for the n th term. **[4]**
- 13.** A bag contains 6 red, 6 blue and 4 green beads. $P(\text{not green})$? **[3]**
- 14.** Write the equation of the line with gradient 2 passing through (0, -1). **[3]**
- 15.** A triangular plot has base 12 m and perpendicular height 7 m. Find its area. **[3]**
- 16.** Factorise $x^2 - 25$. **[4]**
- 17.** $AB \parallel CD$. A transversal makes an acute angle of 49° with AB . Find the co-interior angle on CD . **[3]**
- 18.** Area of a circle radius 21 cm. Take $\pi = 22/7$. **[4]**
- 19.** Solve $2x + 5 < 17$. **[3]**

- 20.** A trader at Chipadze buys a crate of tomatoes for \$20 and sells for \$28. Percentage profit? **[4]**
- 21.** Find the midpoint of (1, 7) and (8, 8). **[3]**
- 22.** $u = (6 ; 0)$, $v = (1 ; 4)$. Find $3u + v$. **[4]**
- 23.** Factorise $x^2 + 7x + 6$. **[4]**
- 24.** A cylindrical drum has radius 7 cm and height 3 cm. Find its volume. $\pi = 22/7$. **[4]**
- 25.** In right-angled triangle XYZ, $\hat{X} = 60^\circ$ and hypotenuse $XZ = 16$ cm. Find XY, adjacent to 60° . **[4]**
- 26.** A ladder of length 5 m leans against a wall. The foot is 3 m from the wall. How far up the wall does it reach? **[4]**
- 27.** Solve $x + y = 10$ and $x - y = 0$. **[4]**
- 28.** Marks (frequency): 2 (6), 3 (5), 4 (4), 5 (3). Find the mean mark. **[4]**
- 29.** A regular polygon has 5 sides. Find each interior angle. **[4]**
- 30.** Solve $5x + 4 = 39$. **[3]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).